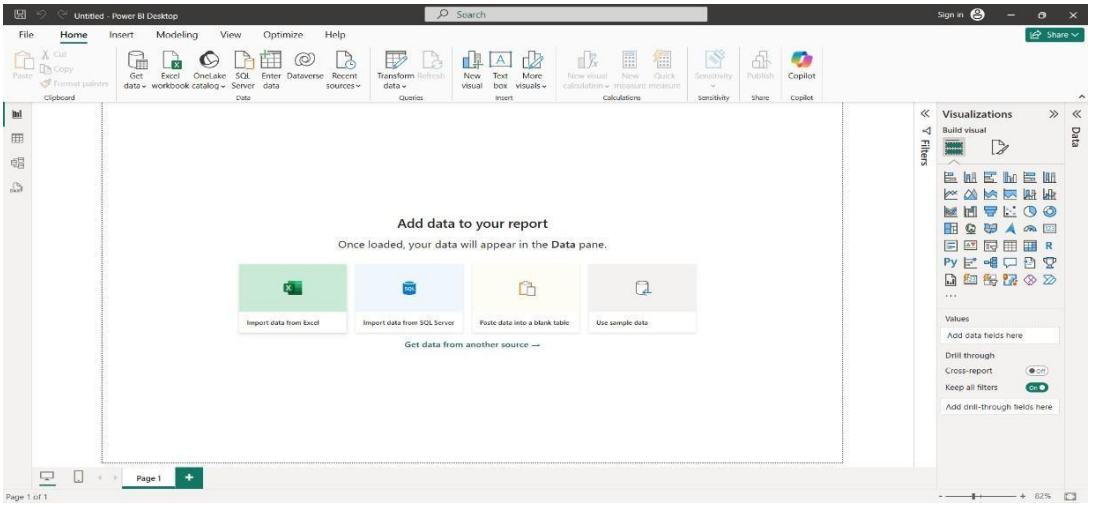
EX.NO: 6	IMPLEMENTATION OF CLASSIFICATION ALGORITHM IN R PROGRAMMING
DATE:	

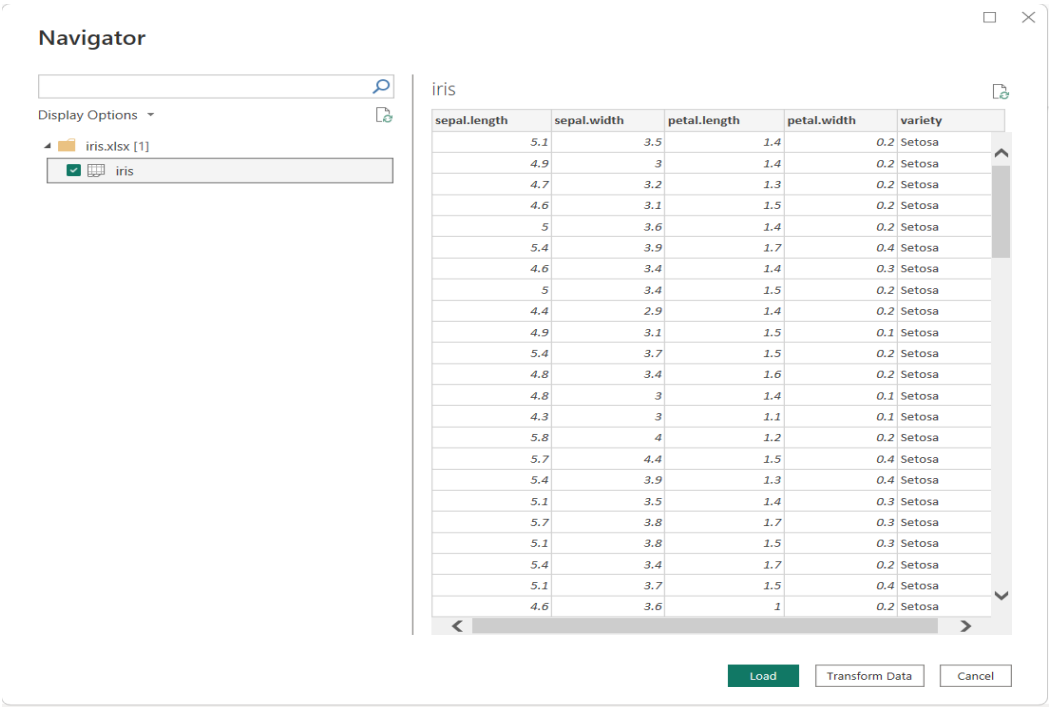
AIM:

To implement the classification algorithm using R programming in Power BI.

PROCEDURE:

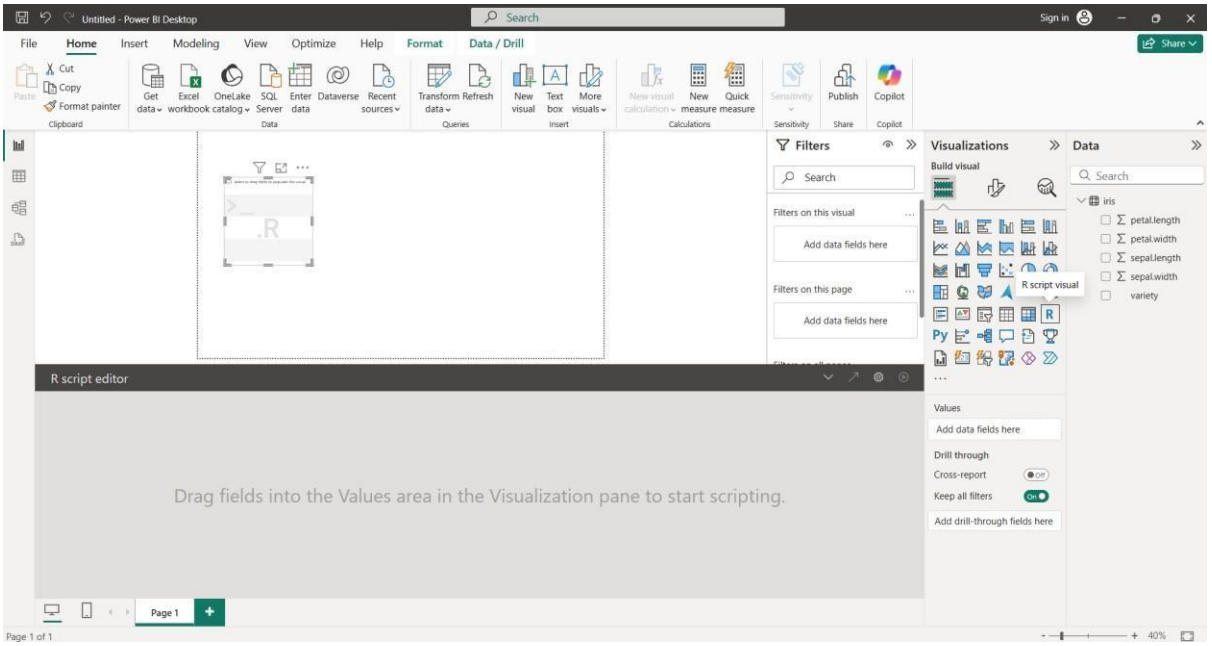
Step 1: Import The Dataset into Power BI



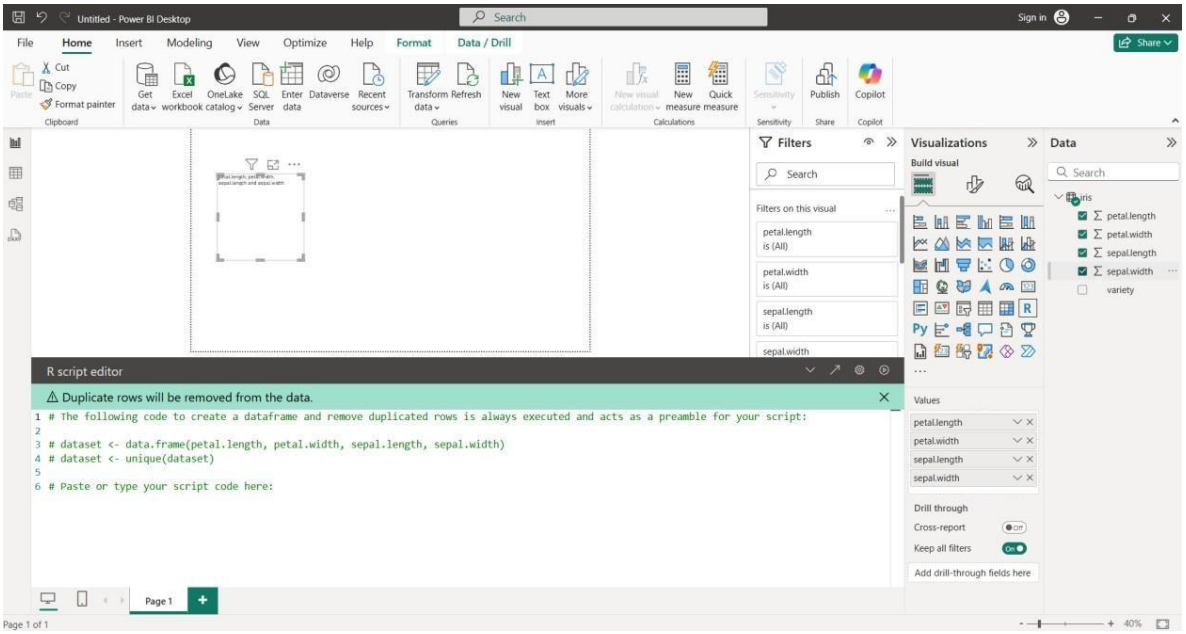


sepal.length	sepal.width	petal.length	petal.width	variety
5.1	3.5	1.4	0.2	Setosa
4.9	3	1.4	0.2	Setosa
4.7	3.2	1.3	0.2	Setosa
4.6	3.1	1.5	0.2	Setosa
5	3.6	1.4	0.2	Setosa
5.4	3.9	1.7	0.4	Setosa
4.6	3.4	1.4	0.3	Setosa
5	3.4	1.5	0.2	Setosa
4.4	2.9	1.4	0.2	Setosa
4.9	3.1	1.5	0.1	Setosa
5.4	3.7	1.5	0.2	Setosa
4.8	3.4	1.6	0.2	Setosa
4.8	3	1.4	0.1	Setosa
4.3	3	1.1	0.1	Setosa
5.8	4	1.2	0.2	Setosa
5.7	4.4	1.5	0.4	Setosa
5.4	3.9	1.3	0.4	Setosa
5.1	3.5	1.4	0.3	Setosa
5.7	3.8	1.7	0.3	Setosa
5.1	3.8	1.5	0.3	Setosa
5.4	3.4	1.7	0.2	Setosa
5.1	3.7	1.5	0.4	Setosa
4.6	3.6	1	0.2	Setosa

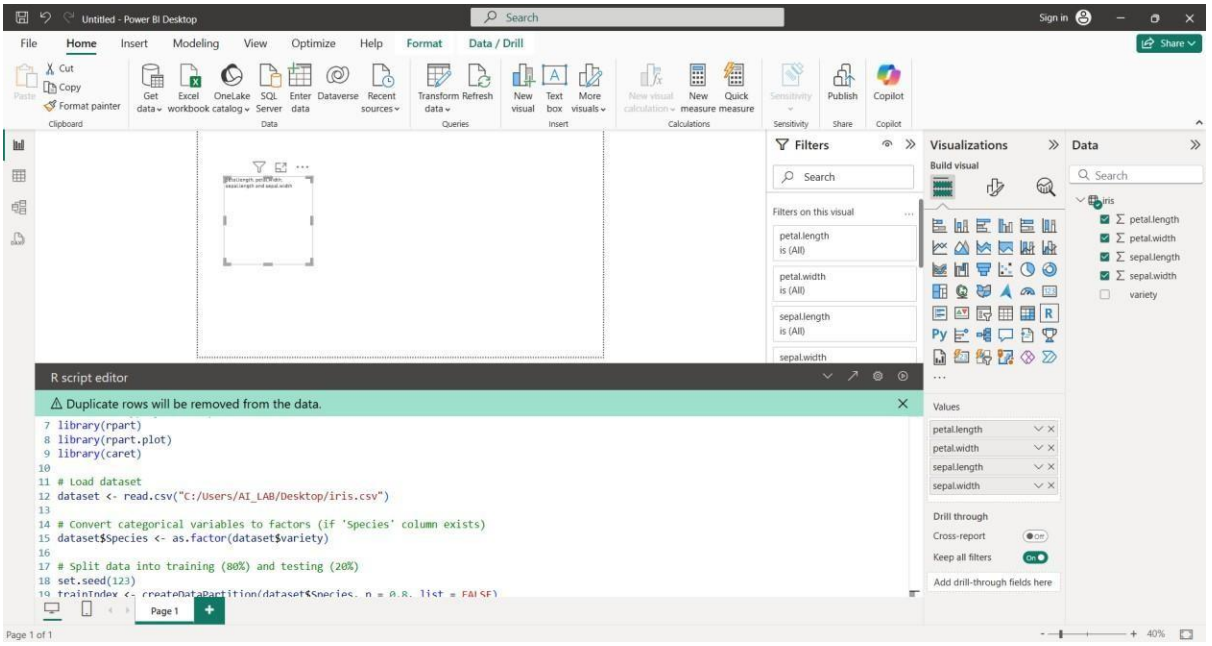
Step 2: Then select R script visual from the Visualization



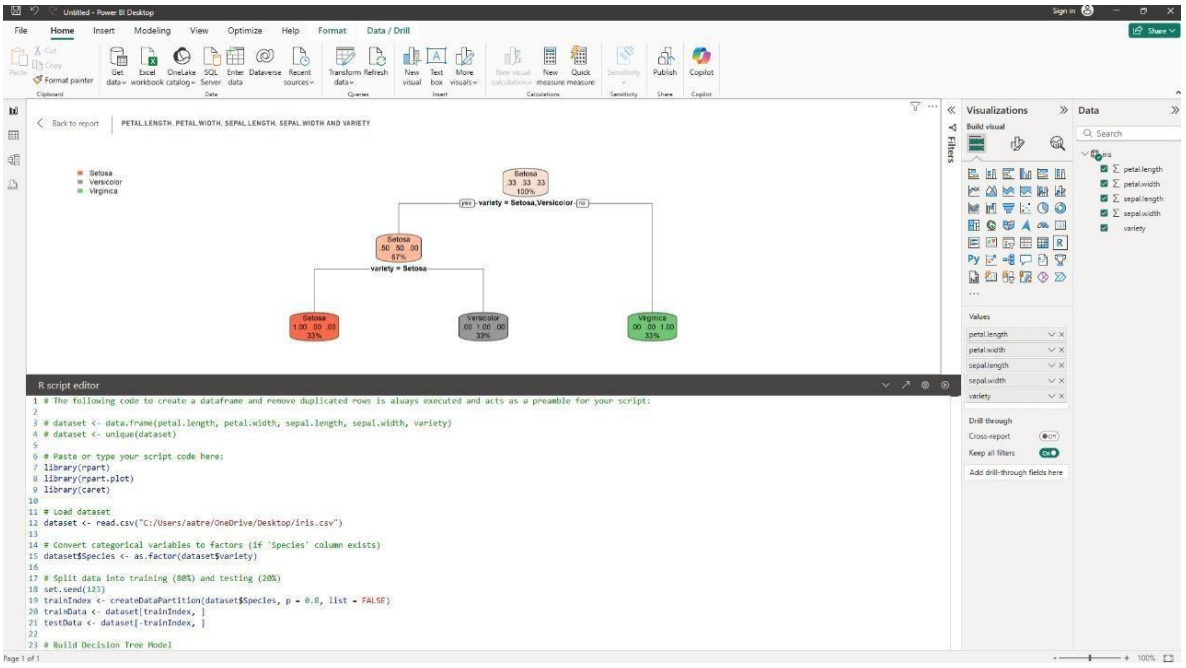
Step 3: Select No of Parameters from the dataset to work within it.



Step 4: In the Script editor Type the Classification algorithm using R for visualization



Step 5: Run the script for the visualization of the code

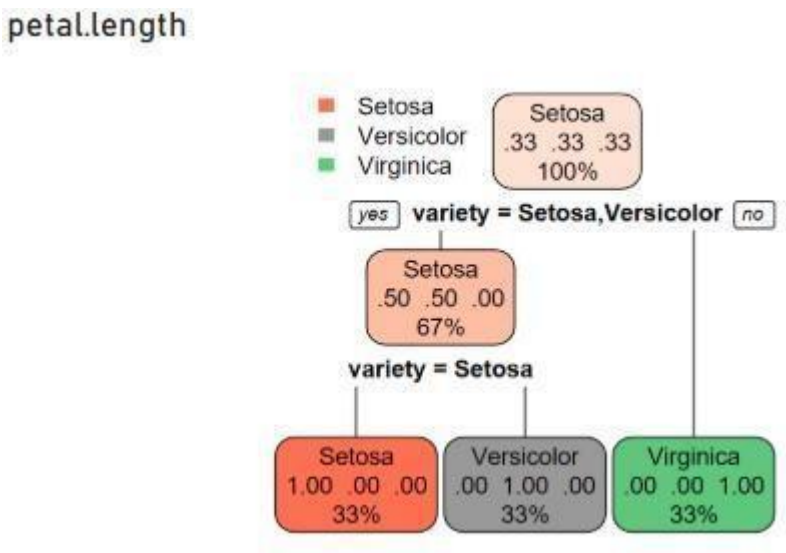


CODE:

```
library(rpart)
library(rpart.plot)
library(caret)
# Load dataset

dataset <- read.csv("C:/Users/AI_LAB/Desktop/iris.csv")
# Convert categorical variables to factors (if 'Species' column exists)
dataset$Species <- as.factor(dataset$variety)
# Split data into training (80%) and testing (20%)
set.seed(123)
trainIndex <- createDataPartition(dataset$Species, p = 0.8, list = FALSE)
trainData <- dataset[trainIndex, ]
testData <- dataset[-trainIndex, ]
# Build Decision Tree Model
dt_model <- rpart(Species ~ ., data = trainData, method = "class")
# Plot the Decision Tree
rpart.plot(dt_model)
# Predict on test data
predictions <- predict(dt_model, testData, type = "class")
# Evaluate model performance
confMatrix <- confusionMatrix(predictions, testData$Species)
print(confMatrix)
```

OUTPUT:



RESULT:

Thus, the implementation of classification algorithm using Decision Tree using R programming in Power BI was executed successfully and the output also verified.